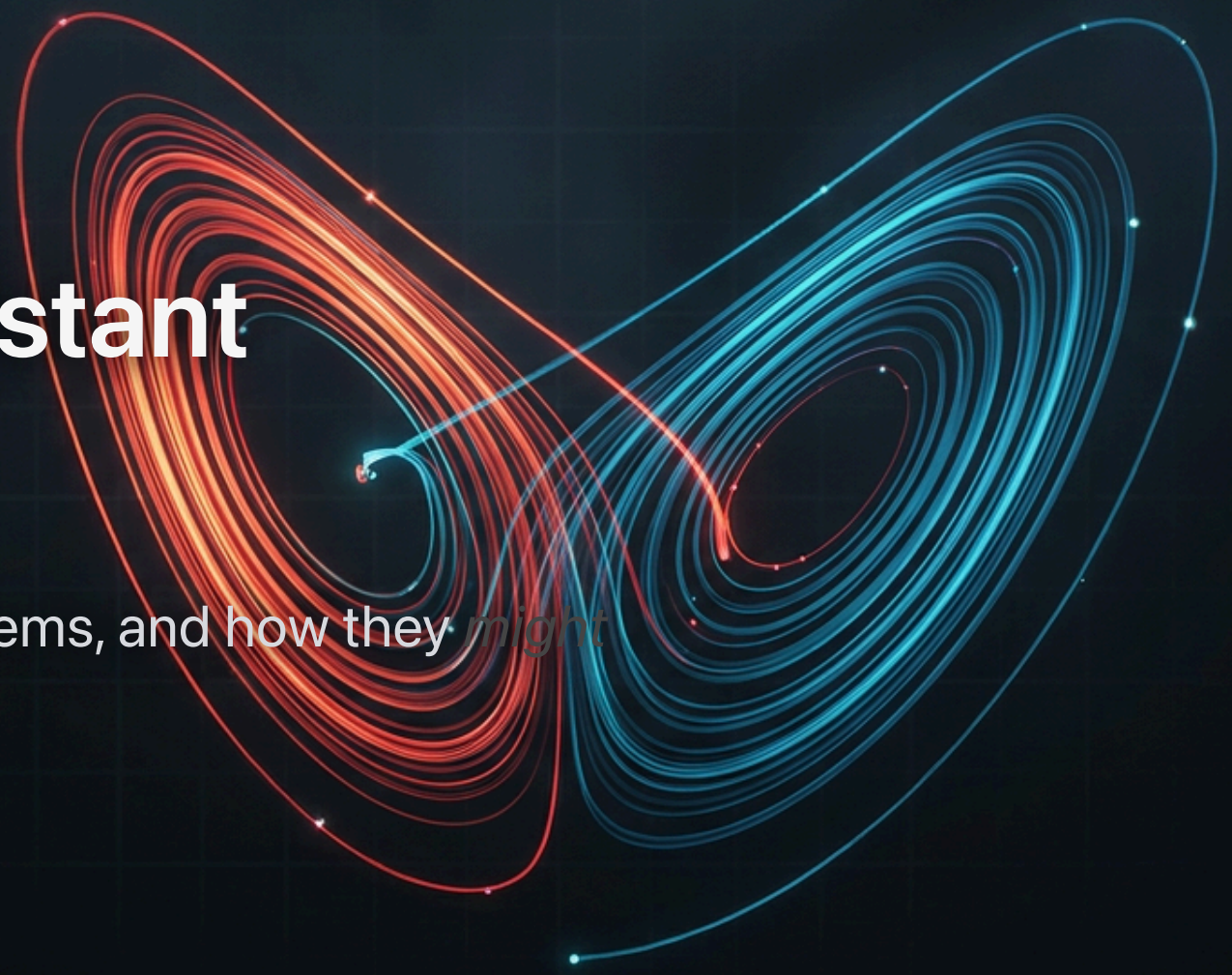
Nearby Prompts, Distant Trajectories

Teaching a lens: chaos, dynamical systems, and how they *might* apply to LLMs.



What I'm not claiming.

- **Not** "LLMs are chaotic." Classical chaos needs things LLMs don't have.
- **Not** "I measured a Lyapunov exponent." Token space is discrete.
- **Not** "bigger = more stable" or "reasoning = stable." Neither holds up.
- **Not** "sentence-embedding distance is ground truth." It's a proxy.
- **Not** "lower divergence = better." Stability is a property, not a score.

> *Setting expectations. This is a Learning Club talk. I'm teaching a lens, not defending a paper.*

What I am claiming.

- LLMs aren't textbook chaotic systems — but at inference time they're **hybrid sequential systems:** continuous activations feed a **discrete branching process.**
- Small semantic or formatting changes can **move distributions** or **flip argmax branches.** The sensitivity is real, and **it varies a lot by model, prompt, and metric.**
- The naive way of measuring this has **specific failure modes** worth naming.
- Chaos vocabulary helps **organize** the phenomenon. It doesn't *prove* anything about LLMs.

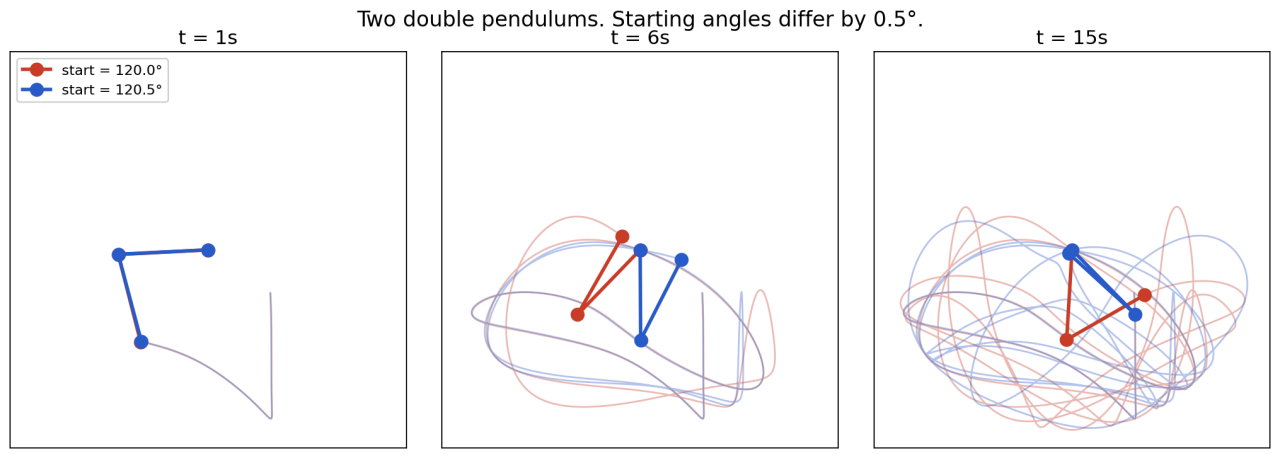
> ***Upshot:*** *treat prompting like operating a high-gain branching system.*
> *Test neighborhoods, not single prompts.*

Chaos isn't randomness.

Chaos is deterministic amplification of small differences.

- Same equations. Starting angles differ by **half a degree**.
- A few seconds later — totally different places.
- Not rolling dice. Magnifying what's already there.

> *Weather forecasts aren't random. They're **sensitive**.*



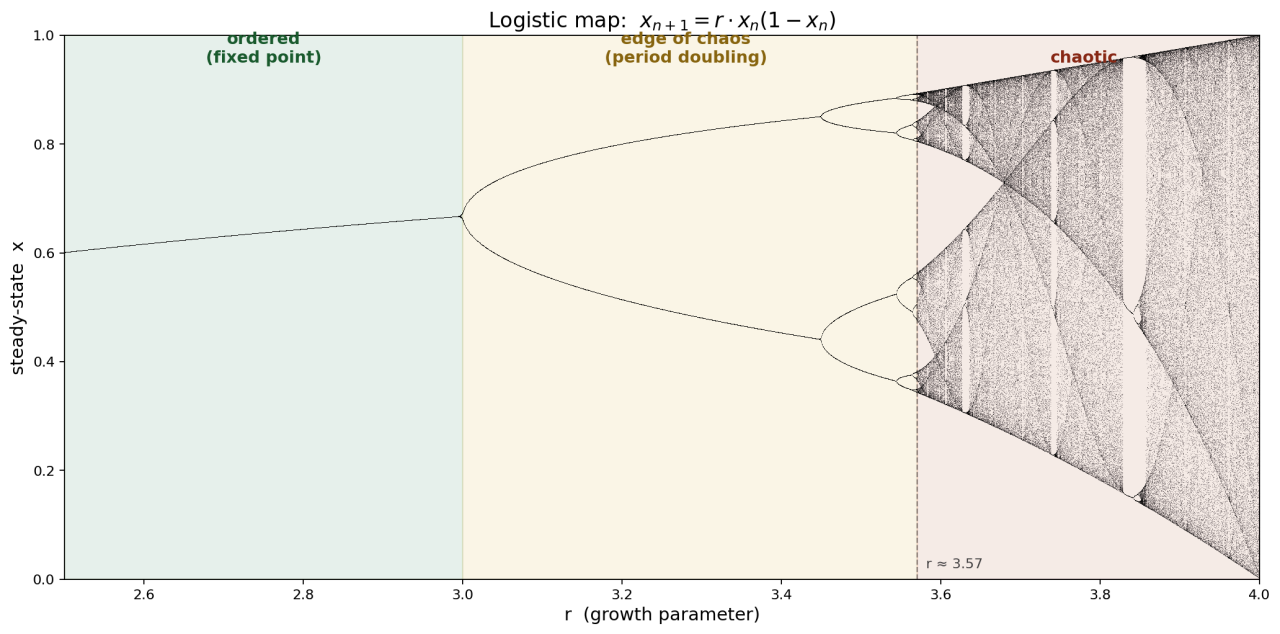
Small differences grow. Measurably.

Logistic map: $x_{n+1} = r \cdot x_n(1 - x_n)$

- **Low r:** single value. **Mid r:** 2, 4, 8 cycles. **High r:** never repeats.

> ***Lyapunov λ*** *— how fast nearby trajectories separate. λ > 0: chaotic.*

Trained nets **sit near this boundary** (Langton 1990; Zhang 2024). *Which side is any LLM on?*



Same input. Same weights. Different output.

Prompt A: `Write a concise Python function that checks whether a string is a palindrome.`

Prompt B: same prompt, **trailing space added**. *(argmax decode — no sampling.)*

Output A — OLMo-3 7B

```
def is_palindrome(s: str) -> bool:
    """
    Check if the given string
    is a palindrome, ignoring
    case and non-alphanumeric
    characters.
    ...
    """
    cleaned = ''.
```

Output B — OLMo-3 7B

```
Certainly! Here's a concise
Python function to check if a
string is a palindrome:
```

```
def is_palindrome(s: str):
    return s == s[::-1]
```

```
How it works: ...
```

This isn't just temperature.

	Same prompt	Tiny prompt change
Temp = 0 (argmax)	Byte-identical. Boring.	★ **this is what the talk measures**
Temp > 0 (sampling)	Different draws, same vibe.	Different draws **and** different vibe. Confounded.

> ***Temperature:*** *from a fixed distribution, what token do we sample?*
> ***Sensitivity:*** *how far did the distribution itself move?*

Starred cell = our probe: zero sampling noise, output still moves. The model's response *function* shifted.

(At T=0.7 on OLMo-3, within-prompt and between-prompt sampling distances can match in magnitude — deterministic decode is the only setting where the signal isn't drowned out.)

So: is an LLM a dynamical system?

- It has **state** (hidden activations, logits, KV cache, prefix).
- It has **iteration** (each token feeds into the next).
- It's **deterministic** under argmax.
- Small input perturbations can produce large output changes.

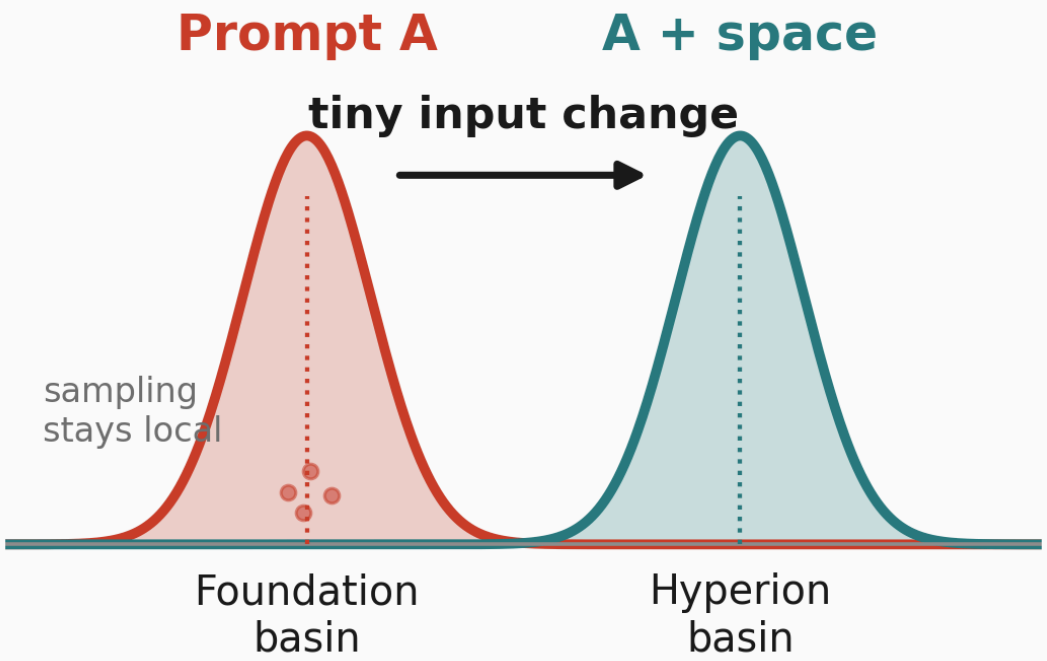
That's the checklist. The remaining question is whether the *magnitude* of amplification is interesting — and whether we can measure it.

> *But there's a catch: classical chaos needs perturbations going to zero. Token space is discrete. We'll come back to this.*

Both outputs can be correct.

> *A double pendulum isn't "wrong"; it obeys physics and lands elsewhere. Same bar for LLMs.*

- "Book like *Dune*" → Foundation. Add a trailing space → Hyperion.
- Both recommendations are defensible; neither is a hallucination.
- Sampling stays local; sensitivity can **move the distribution**.
- **Measure:** divergence per unit of *meaning-preserving* input change.



State, and prior work — short version

An LLM's "state" is hidden activations + logits + prefix + KV cache. *How much does the output distribution move when the input changes a little?*

People have started probing this:

- **Li et al. 2025** — QLE-style analysis on Qwen2-14B. Magnitudes grow ~1.32× per layer; shallow layers contract, deep layers amplify. They call it *quasi*-Lyapunov (finite depth, not a fixed map).
- **Geshkovski et al. 2023** — self-attention as interacting-particle dynamics.
- **Poole / Schoenholz** — edge-of-chaos signal propagation in deep nets.

> *The chaos math is cleanest in **activation space**.*
> *What I measure is the downstream shadow at the **output-text level**.*

The experiment

- **~18 models:** Qwen (0.8B → 9B), Gemma 4, Phi-4, DeepSeek-R1, Mistral, Granite, Falcon, SmolLM, OLMo 2 & 3; legacy: GPT-2 XL, GPT-J, Pythia, OPT, LLaMA-1.
- **Prompt ladder:** identical / no-op formatting / punctuation / synonym / paraphrase / small semantic / positive control.
- **Deterministic decode** (`do_sample=False`, argmax) — divergence is a shift in the model's most confident response.
- **Metrics:** sentence-embedding cosine distance (primary) + token edit + hidden-state distance + logit KL. All proxies; no ground truth.
- **Analysis:** bootstrap CIs + paired permutation tests. Present clusters, not ranks.
- **Reproducibility:** pinned HF revisions, bf16, chat templates disabled, config published with artifacts.

Within-Qwen: one clean contrast.

n = 24 prompt pairs, same family (controls for pretraining + architecture):

Model	Mean	95% CI	vs 4B (p)
Qwen3.5 **4B**	0.034	[0.018, 0.053]	—
Qwen3.5 **9B**	0.037	[0.016, 0.061]	0.78
Qwen3.5 **2B**	0.073	[0.039, 0.115]	**0.012**
Qwen3.5 **0.8B**	0.089	[0.048, 0.137]	**<0.001**

- 0.8B is **meaningfully more sensitive** than 4B within the same family.
- 4B vs 9B: indistinguishable at this n — **not claiming "bigger = stable."**
- Caveat: 4B/9B emit `Thinking Process:` preambles. Scaffold confound, next slide.

Scaffold "stability" is mostly metric artifact.

Short outputs (64 tokens): scaffolded models look ~4× more stable. Identical `<think>` preambles dominate sentence-embedding similarity.

Warning about evaluation — not a claim about model dynamics.

Long outputs (512 tokens) expose the mixed bag:

Scaffolded model	512-tok semantic	Prompt-end top-1 prob
DeepSeek-R1 7B	0.027 (stable)	0.99976
Qwen 4B / 9B	0.050 / 0.057	0.970 / 0.988
SmolLM3 3B	0.080 (middle)	0.99983
Phi-4 reasoning+	**0.160 (brittle)**	**0.99999996**

Phi-4: certain at the prompt, chaotic at 512 tokens. Scaffold adherence ≠ content robustness.

Qwen thinking-off control: small effect, not monotonic — scaffold helps big Qwens a bit, *hurts* Qwen 0.8B.

Era, recipe, and the LLaMA-1 surprise

512-token semantic distance, non-scaffold models only:

Model	Semantic	Era
LLaMA-1 7B	**0.053**	2023 base — stable outlier
Gemma E2B **instruct**	0.056	modern chat
Mistral 7B v0.3	0.068	modern chat
Gemma E4B **instruct**	0.072	modern chat
Gemma E4B **base**	0.119	modern base
Gemma E2B **base**	0.199	modern base
GPT-2 XL / OPT / Pythia / GPT-J	0.14 – 0.22	pre-chat base

LLaMA-1 is content-stable without a scaffold. Within Gemma, **instruct ≫ base** — recipe, not calendar. Era alone does not predict sensitivity; token-path and semantic metrics can even point different ways.

Stability isn't responsiveness.

> *A model that outputs `"the the the"` regardless of input is extremely stable. So is a model collapsed onto one fixed answer.* ***Neither is what we want.***

One thing I tried: **sweep Qwen through BF16 / 8-bit / 4-bit quantization.** Qwen 0.8B at 4-bit scored *lower* perturbation divergence (0.138 → 0.091) — which sounds like "more stable" until you check its drift from BF16 on identical prompts (**0.132 — huge**).

Consistent with collapse onto a narrower output manifold, not robustness. A naive stability metric **can't tell a robust model from a collapsed one.**

> ***Fix:*** *always pair perturbation distance with drift-from-baseline. Both axes, not one.*

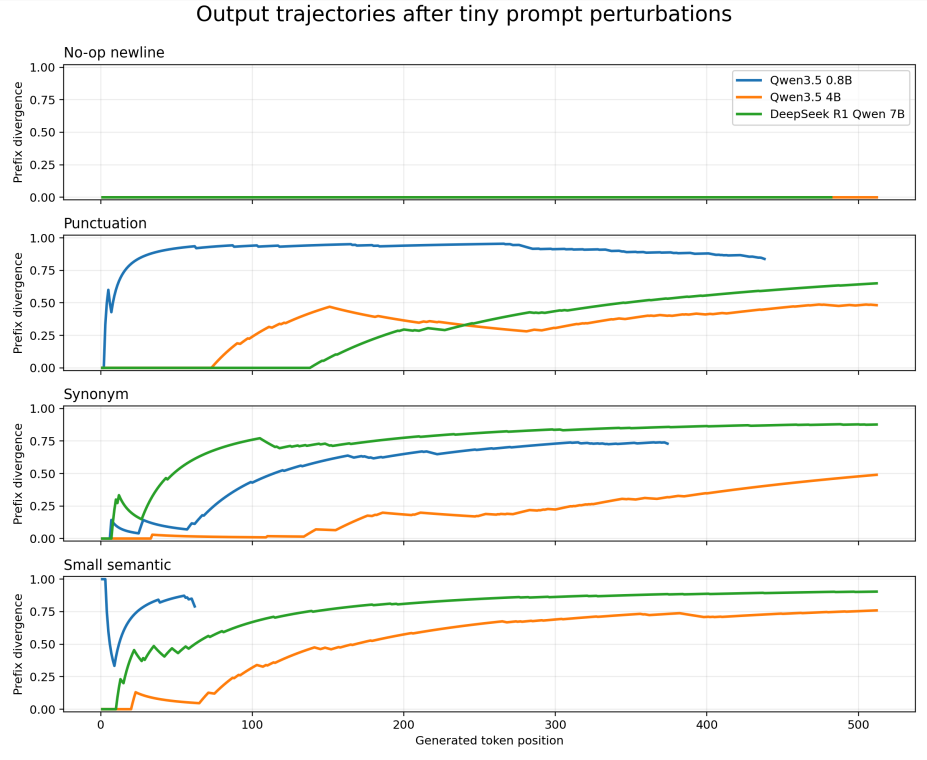
Measuring is the hard part.

Three ways a naive stability probe will mislead you:

Confound	What happens	Caught by
Collapse	Degenerate model scores "stable" because outputs stop responding to input. (Qwen 0.8B 4-bit.)	Distance from baseline on identical prompts.
Scaffold	Short-output score dominated by deterministic preamble. (Qwen 4B/9B, SmolLM3.)	Longer continuations; scaffold stripping.
Confident ≠ stable	Low prompt-end JS + sharp top-1 argmax doesn't mean trajectory is stable. **Phi-4 is certain at prompt-end and brittle at 512 tokens.**	Multi-scale measurement: short vs long, logit vs text.

> *The most useful contribution of this work isn't the numbers.*
> ***It's naming the failure modes before the field starts quoting the numbers.***

Long-generation trajectories



Early growth, then saturation as outputs enter different textual basins. DeepSeek-R1 keeps a 138-token shared prefix on a punctuation-only perturbation.

Mechanism: boundary beats bulk

Cross-model correlations with 512-token semantic divergence (n=20):

Logit signal at prompt-end	Pearson r	Reads as
Top-1 probability	**−0.84**	more confident next-token → less downstream drift
Top-1 flip rate	+0.57	argmax crossings predict later divergence
Top-1 margin (logit)	−0.39	weaker but same direction
Full-vocab JS divergence	−0.10	**bulk distribution shift doesn't predict anything**

> *Small prompt change → argmax crosses a low-margin boundary → different first token → autoregressive feedback → different trajectory.*

The distribution often isn't moving much in bulk. It's that a low-margin next-token decision is fragile, and one flipped argmax steers the generation into a different basin.

A question the lens suggests (not a claim).

Static floor

How few bits to store the model?

- TurboQuant, KIVI, KV quantization.
- Rate-distortion bounds.
- Well-characterized.

Dynamical floor?

How few bits before *behavior* drifts?

- Might depend on model sensitivity.
- Stable models *might* tolerate more compression.
- **Open. My data doesn't settle it.**

> *Compression has a static floor. Does it have a dynamical one too?*
> *The chaos lens suggests the question. I don't have the answer.*

The practitioner upshot.

> ***Don't evaluate on a single prompt, single decode, or single metric. Prompting is operating a high-gain branching system.***

Operational:

- **Reliability:** test prompt *neighborhoods*, not one canonical prompt.
- **Model comparison:** report sensitivity *ranges* over equivalent prompts.
- **Output metrics:** strip boilerplate, compare answer spans, watch prefixes.
- **Decoding:** deterministic for sensitivity; sampling separately for deployment.
- **Quantization:** lower divergence ≠ robustness. Also check baseline drift.

> ***The chaos lens doesn't prove anything about LLMs. It suggests questions that benchmarks don't ask — and those questions are worth asking.***

Questions?

Backup — "Would I get the same answer if I ran it?"

Argmax decode has no sampling step. `do_sample=False` → highest-logit token wins each step. Seed is inert.

- Same prompt twice → byte-identical output.
- Prompt A vs B → top-token at some position **flipped**. Most confident response moved.

Temperature > 0? 30-sample cluster test (OLMo-3, palindrome pair, T=0.1):

- Prompt A: 30 samples cluster tightly. Prompt B: same.
- A-cluster and B-cluster are visibly separate.

Not "got unlucky once" — two different attractors. Sampling noise smaller than the shift between them.

Backup — "Is this chaos?" defense

- Formal chaos needs exponential divergence under iteration. **Not proven.**
- What was measured: small input perturbations producing different outputs, varying by model, reproducible under deterministic decode.
- Consistent with behavior near a chaos boundary. Not a proof of chaos.
- The frame is the contribution; the experiment is a probe, not a theorem.

Backup — Related work I came across late

Prompt sensitivity / brittleness:

- **Salinas & Morstatter 2024** — "Butterfly Effect of Altering Prompts." My whitespace example, already published.
- **Sclar et al. 2023** — formatting sensitivity; up to 76-point swings on LLaMA-2-13B.
- **Lu et al. 2021** — example ordering alone moves few-shot near-random to near-SOTA.
- **PromptRobust / POSIX / RobustAlpacaEval** — published sensitivity benchmarks.

Dynamical systems in NNs:

- **Poole 2016 / Schoenholz 2017** — edge-of-chaos signal propagation.
- **Geshkovski et al. 2023** — attention as interacting-particle dynamics.
- **Tomihari & Karakida 2025** — Jacobian/Lyapunov analysis of self-attention.

Backup — Statistical honesty

- **n = 9** prompt pairs per model; **n = 24** in hardened Qwen wave. Small.
- **Robust at n = 24:** Qwen 4B vs 0.8B $p<0.001$; Qwen 4B vs 2B $p=0.012$; cluster membership.
- **Not robust at this n:** Qwen 4B vs 9B ($p=0.78$); middle-pack ordering; standalone quant flip.
- Scaffold vs non-scaffold is **confounded with post-training recipe** — needs different *models*, not more prompts.

> *If you want to disagree with "stable cluster vs brittle cluster" you need 100+ prompts. If you want to disagree with specific model orderings — you're already right, I'm not claiming them.*

Backup — Failed experiments

- **gpt-oss-20b:** MXFP4 / Triton driver mismatch on SageMaker image.
- **Nemotron Nano 9B v2:** container lacked `mamba-ssm`.
- **Phi-4 mini:** Transformers version / custom-code import failure.

Reported as tooling misses, not stability findings.

Backup — Full bootstrap readout (512 tokens)

Stable / mid

Model	Mean	95% CI
DeepSeek-R1 Qwen 7B	0.027	0.018 – 0.036
Qwen3.5 4B	0.050	0.033 – 0.066
LLaMA-1 7B	0.053	0.017 – 0.100
Gemma 4 E2B it	0.056	0.033 – 0.080
Qwen3.5 9B	0.057	0.037 – 0.075
Mistral 7B v0.3	0.068	0.047 – 0.089
Gemma 4 E4B it	0.072	0.038 – 0.110
Qwen3.5 2B	0.075	0.050 – 0.103

Higher sensitivity / caveats

Model	Mean	95% CI
OLMo 2 7B	0.088	0.055 – 0.127
Qwen3.5 0.8B	0.103	0.061 – 0.153
OLMo 3 7B	0.104	0.077 – 0.135
Gemma 4 E4B base	0.119	0.071 – 0.173
GPT-2 XL	0.144	0.082 – 0.208
Phi-4 reasoning+	**0.161**	0.072 – 0.255
Gemma 4 E2B base	0.199	0.152 – 0.248

n=24 pairs, 512 tokens. Clusters, not ranks. **Phi-4: scaffolded yet brittle** — scaffold ≠ stable.